

MA7202 Introduction to Functional Data Analysis 2025-26

Mini Project

DECLARATION

All sentences or passages quoted in this project thesis from other people's work have been specifically acknowledged by clear cross referencing to author, work and page(s). I understand that failure to do this amounts to plagiarism and will be considered grounds for failure in this module and the degree examination as a whole.

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Date: 29/04/2026

Introduction

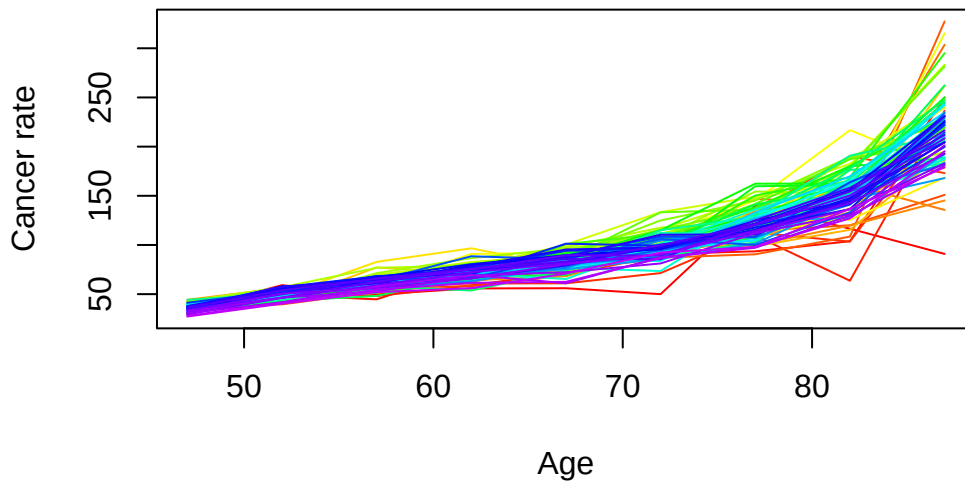
Breast cancer is Australia's most prevalent female cancer, representing approximately 28% of female cancer diagnoses (AIHW, 2023). Age-specific incidence rises steeply from midlife onward driven by cumulative hormonal exposure and somatic mutation accumulation (Pike et al., 1983).

This report applies Functional Data Analysis (FDA) to the Cancerrate dataset which records age-specific rates for nine age groups across 81 years from 1921 to 2001, treating each year's observations as a smooth continuous function of age rather than isolated measurements (Ramsay

& Silverman, 2005). Three analytical stages are undertaken: functional smoothing, descriptive statistics and functional linear regression analysis coupled with permutation hypothesis testing.

```
library(fda)

data(Cancerrate)
plot(Cancerrate)
```



```
cancer_data <- Cancerrate$y
```

Functional smoothing

The raw plot of the Cancerrate data shows a fan shape and a divergence at older ages. Since breast cancer rates increase with age and exhibit no periodic structure a B-spline basis is the choice (Ramsay & Silverman, 2005). A fourth-order cubic B-spline with $n_{\text{basis}} = 11$ provides a saturated representation of the nine observations per curve with continuous second derivatives. The roughness penalty λ penalises curvature, selected objectively via Generalised Cross-Validation (GCV).

```

age      = seq(47, 87, 5)
ageRng   = c(45, 90)
norder   = 4
nbasis   = length(age) + norder - 2
cancer_basis = create.bspline.basis(ageRng, nbasis, norder)
harmacellfd = int2Lfd(max(0, norder - 2))

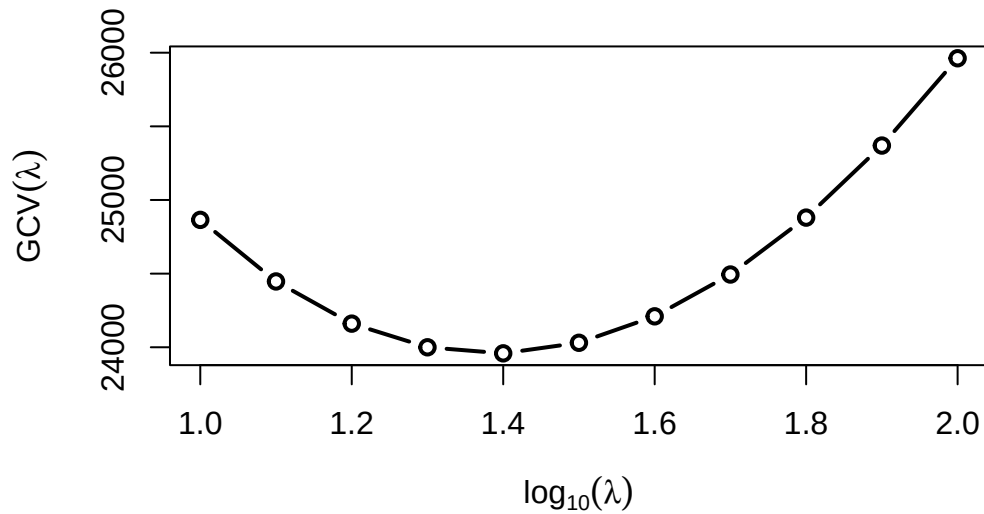
loglam   = seq(1, 2, 0.1)
nlam     = length(loglam)
gcvsave  = rep(NA, nlam)

for (ilam in 1:nlam) {
  lambda      = 10^loglam[ilam]
  cancer_fdPar = fdPar(cancer_basis, harmacellfd, lambda)
  smooth_list  = smooth.basis(age, cancer_data, cancer_fdPar)
  gcvsave[ilam] = sum(smooth_list$gcv)
}

plot(loglam, gcvsave, type = 'b', lwd = 2,
     xlab = expression(log[10](lambda)),
     ylab = expression(GCV(lambda)),
     main = "GCV Criterion for Breast Cancer Rate Smoothing")

```

GCV Criterion for Breast Cancer Rate Smoothing



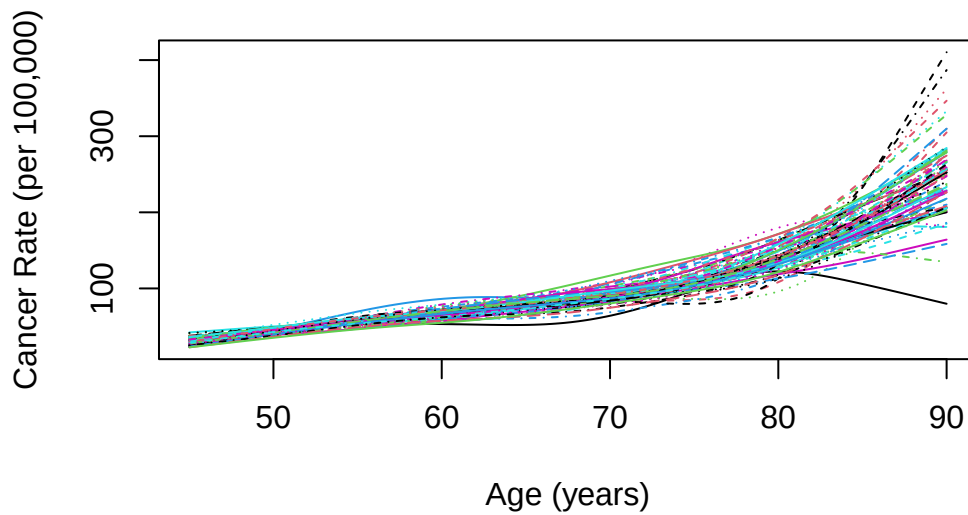
The GCV minimum at $\log_{10}(\lambda) = 1.4$ ($\lambda \approx 25.12$) removes noise while preserving the exponential age-trend. All 81 smoothed curves share the same broadly exponential shape, rising steeply from around 30 per 100,000 at age 47 to over 250 per 100,000 by age 87.

```
best_lambda = 10^1.4
cancer_fdPar_final = fdPar(cancer_basis, harmaccelLfd, best_lambda)
cancer_smooth = smooth.basis(age, cancer_data, cancer_fdPar_final)
cancer_fd = cancer_smooth$fd

cancer_fd$fdnames = list(
  "Age (years)",
  "Year (1921-2001)",
  "Cancer Rate (per 100,000)"
)

plot(cancer_fd, lwd = 1,
     main = "Smoothed Breast Cancer Rates by Age (1921-2001)",
     xlab = "Age (years)",
     ylab = "Cancer Rate (per 100,000)")
```

Smoothed Breast Cancer Rates by Age (1921–2001)

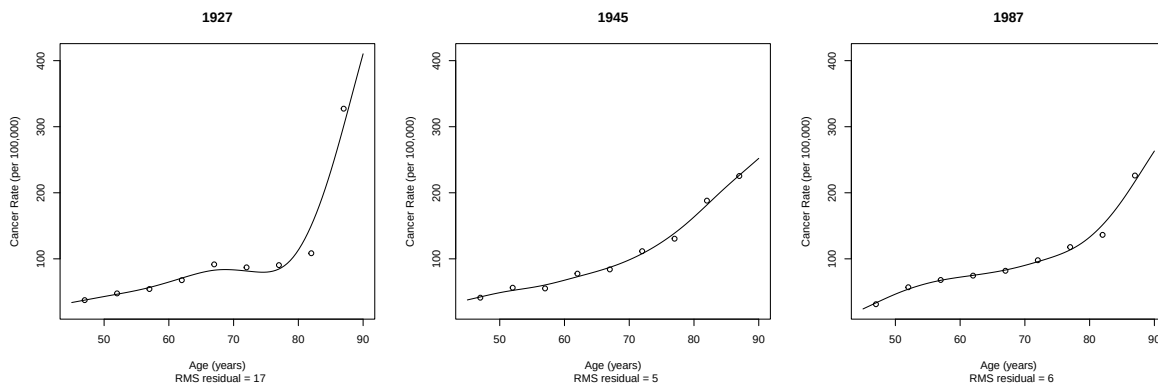


[1] "done"

Individual fit plots for representative years (1927, 1945, 1987) confirm RMS residuals of 17, 5, and 6 respectively and should be acceptable given the scale of the data of up to 300 per 100,000 females.

```
par(mfrow=c(1,3), ask = FALSE)
plotfit.fd(cancer_data,age,cancer_fd, index = c(7,25,67))
```

Multiple plots: Click in the plot to advance to the next plot



```
par(mfrow=c(1, 1))
```

Descriptive statistics

The mean function quantifies a typical year showing rates rising from roughly 35 per 100,000 at age 47 to around 255 per 100,000 at age 87 consistent with the hormonal age-incidence model of Pike et al. (1983). The standard deviation function mirrors this pattern remaining below 10 per 100,000 up to age 70 before increasing sharply to over 50 per 100,000 at age 87. This trend shown to be stable at young ages but highly variable at old ages mirrors findings by McCredie et al. (2004) who documented that Australian late age incidence increased in the second half of the 20th century due to both epidemiological change and improved detection.

```
mean_cancer    = mean.fd(cancer_fd)
std_cancer     = std.fd(cancer_fd)

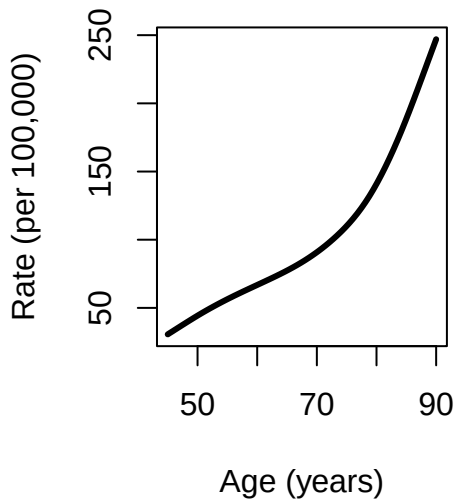
#| fig-width: 20
#| fig-height: 10
par(mfrow = c(1, 2), ask = FALSE)

plot(mean_cancer, lwd = 3,
     main = "Mean Cancer Rate by Age",
     xlab = "Age (years)",
     ylab = "Rate (per 100,000)")
```

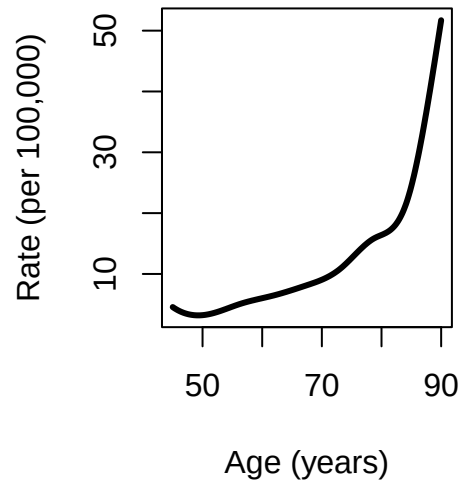
```
[1] "done"
```

```
plot(std_cancer, lwd = 3,
     main = "Std Dev of Cancer Rate",
     xlab = "Age (years)",
     ylab = "Rate (per 100,000)")
```

Mean Cancer Rate by Age



Std Dev of Cancer Rate



[1] "done"

```
par(mfrow = c(1, 1))
```

Research Question

- Does calendar year have a statistically significant effect on the age-specific breast cancer rate profile and at which ages has incidence changed most over the 81-year period?

To explore this, traditional scalar regression is inadequate because it cannot capture the continuous nature of the age-incidence relationship. We rather use a specific functional linear model featuring a functional response and a scalar independent variable. This method is mathematically motivated because the predictor calendar year is a single scalar value for each observation while the response is a continuous curve representing the cancer rate across ages. The model is represented as:

$$\text{CancerRate}_i(t) = \beta_0(t) + \text{Year}_i \cdot \beta_1(t) + \varepsilon_i(t)$$

Where $\beta_0(t)$ is the baseline intercept function, $\beta_1(t)$ is the regression coefficient function capturing the effect of the year across different ages and $\varepsilon_i(t)$ represents the residual variation.

A Permutation F-test is applied and it then allows us to test the null hypothesis ($H_0 : \beta_1(t) = 0$) for all t against the alternative hypothesis ($H_1 : \beta_1(t) \neq 0$) for some t , evaluating whether the year has a statistically significant impact on the age-specific cancer rate profile across the 81-year period.

```
year_scalar <- 1921:2001

xfdlist <- list(const = rep(1, 81),
               year  = as.numeric(year_scalar))

betafdPar <- fdPar(cancer_basis, harmaccelLfd)
betalist  <- list(const = betafdPar,
               year  = betafdPar)

cancerRegress <- fRegress(cancer_fd, xfdlist, betalist)
betaestlist <- cancerRegress$betaestlist
cancer_fine  <- seq(45, 90, length.out = 200)
```

The first panel below shows $\beta_0(t)$ the intercept function with very wide confidence bands that widen above age 80 and reflect high uncertainty in the baseline which is consistent with the high variability already identified in the standard deviation function.

The panel under shows the year coefficient function $\beta_1(t)$ and this is the critical plot for answering the research question. The estimated coefficient sits close to zero across almost the entire age range from 47 to 85. The 95% confidence bands cross the red zero dashed line throughout the whole domain, meaning that at no age with the year effect statistically distinguishable from zero with confidence. Between ages 50 and 80 the coefficient oscillates slightly around 0.0 to 0.05 suggesting at most a marginal increase of around 0.05 per 100,000 per calendar year at some ages. At the upper end around age 85–90, the coefficient turns sharply negative and the confidence band widens but this is driven by data sparsity at the oldest age group rather than a genuine signal. The overall picture from this panel is that calendar year which is modelled as a linear scalar predictor does not produce a statistically significant or stable effect on the cancer rate profile at any individual age.

```
cancer_hatfd <- cancerRegress$yhatfdobj
cancer_hatmat <- eval.fd(age, cancer_hatfd)
resmat       <- cancer_data - cancer_hatmat
SigmaE       <- cov(t(resmat))

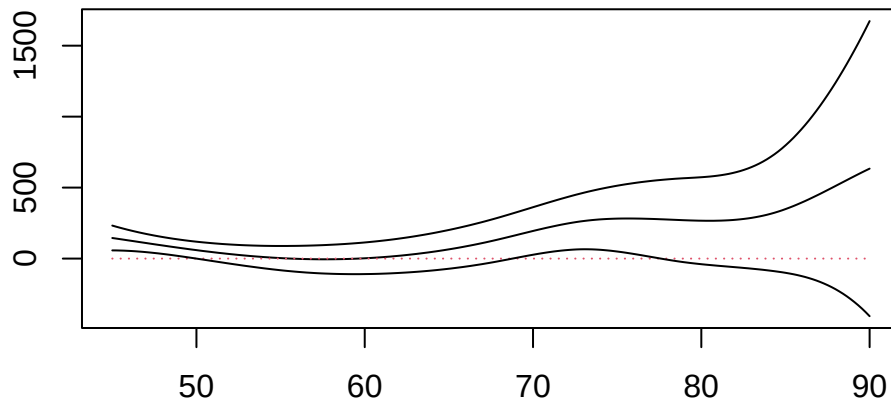
y2cMap <- cancer_smooth$y2cMap

cancerRegressSE <- fRegress(cancer_fd, xfdlist, betalist,
                          y2cMap = y2cMap)
```

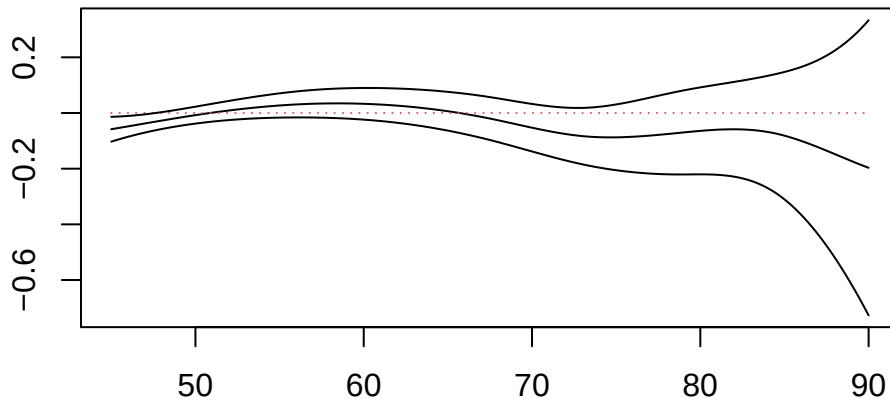
```
cancerRegressErr  <- fRegress.stderr(cancerRegressSE, y2cMap, SigmaE)
betastderrlist    <- cancerRegressErr$betastderrlist

plotbeta(betaestlist, betastderrlist, cancer_fine)
```

Regression function 1



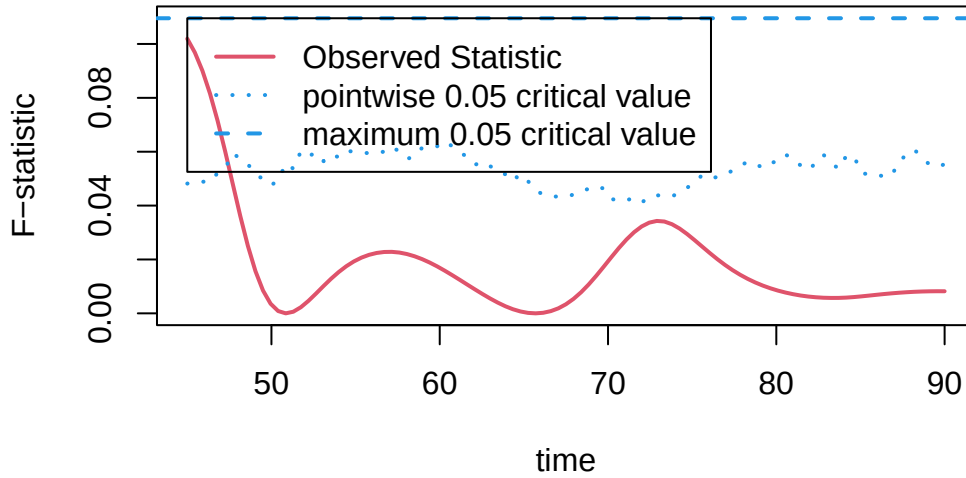
Regression function 2



The observed F-statistic represented by the red solid line begins at approximately 0.08 at age 47 which is the highest point in the plot and then drops below the 0.05 pointwise critical value at approximately 0.045–0.055 by around age 50. It remains below it for the entire remaining age range from 50 to 90. Two small secondary peaks appear around ages 57 and 72 but neither exceed the critical value. The observed statistic never approaches the maximum 0.05 critical value at any age, which is the threshold accounting for multiple testing across the functional domain. The only region where there is any signal is around age 47–49, where the observed statistic is above the threshold but still below the maximum threshold and thus it is not sufficient to reject the null hypothesis.

```
set.seed(123)
F.res <- Fperm.fd(cancer_fd, xfdlist, betalists)
```

Permutation F-Test



Conclusion

The permutation F-test fails to reject the null hypothesis as calendar year as a linear scalar predictor does not have a statistically significant effect on the age-specific breast cancer rate profile across the 81-year period from 1921 to 2001. This finding does not mean rates did not change but rather that the descriptive analysis clearly showed they did more at older ages. What the functional linear regression analysis reveals is that the change was not a simple linear year on year trend across the full age profile. The changes were rather concentrated in specific periods and in specific age groups producing a non-linear temporal pattern that a linear year predictor cannot capture.

References

- Australian Institute of Health and Welfare (AIHW) (2023). *Cancer data in Australia*. Available at: <https://www.aihw.gov.au/reports-statistics/health-conditions-disability-deaths/cancer/data>
- McCredie, M., Coates, M., Grulich, A. and Armstrong, B. (2004). Breast cancer incidence in Australian women aged 40 years and over. *British Journal of Cancer*, 90(7), pp. 1377–1381.

Pike, M.C., Krailo, M.D., Henderson, B.E., Casagrande, J.T. and Hoel, D.G. (1983). 'Hormonal' risk factors, 'breast tissue age' and the age-incidence of breast cancer. *Nature*, 303(5920), pp. 767–770.

Ramsay, J.O. and Silverman, B.W. (2005). *Functional Data Analysis* (2nd ed.). Springer, New York.